

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Uses sum of probabilities = 1 to correctly find k	AO1.1b	B1	$2k + 4k + 6k + 8k = 1$ $20k = 1$
Calculates $E(Y)$ or $E(Y^2)$ correctly	AO1.1a	M1	$k = \frac{1}{20}$ or 0.05
Calculates both $E(Y)$ and $E(Y^2)$ correctly	AO1.1b	A1	$E(Y) = 1 \times 2k + 2 \times 4k + 3 \times 6k + 4 \times 8k$ $= 60k$ or 3
Calculates $\text{Var}(Y)$	AO1.1a	M1	$E(Y^2) = 1^2 \times 2k + 2^2 \times 4k + 3^2 \times 6k + 4^2 \times 8k$ $= 200k$ or 10
Uses formula for $\text{Var}(aY \pm b)$	AO1.1a	M1	$\text{Var}(Y) = E(Y^2) - E(Y)^2$ $= 200k - (60k)^2$ or $10 - 3^2$ $= 200k - 3600k^2$ or 1
Completes a fully correct, logical argument to show that $\text{Var}(5Y - 2) = 25$ AG	AO2.1	R1	$\text{Var}(5Y - 2) = 5^2 \text{Var}(Y)$ $= 5^2 \times 1$ $= 25$

ALT

Marking Instructions	AO	Marks	Typical Solution										
Uses sum of probabilities = 1 to correctly find k	AO1.1b	B1	$2k + 4k + 6k + 8k = 1$ $20k = 1$										
Finds values of $5Y - 2$ for each value of y	AO1.1a	M1	$k = \frac{1}{20}$ or 0.05 <table><tr><td>y</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>$5y - 2$</td><td>3</td><td>8</td><td>13</td><td>18</td></tr></table>	y	1	2	3	4	$5y - 2$	3	8	13	18
y	1	2	3	4									
$5y - 2$	3	8	13	18									
Calculates $E(5Y - 2)$ or $E((5Y - 2)^2)$ correctly	AO1.1a	M1	$E(5Y - 2) = 3 \times 2k + 8 \times 4k + 13 \times 6k + 18 \times 8k$ $= 260k$ or 13										
Calculates $E(5Y - 2)$ and $E((5Y - 2)^2)$ correctly	AO1.1b	A1	$E((5Y - 2)^2) = 3^2 \times 2k + 8^2 \times 4k + 13^2 \times 6k + 18^2 \times 8k$ $= 3880k$ or 194										
Calculates $\text{Var}(5Y - 2)$	AO1.1a	M1	$\text{Var}(5Y - 2) = E((5Y - 2)^2) - E(5Y - 2)^2$ $= 3880k - (260k)^2$ or $194 - 13^2$ $= 3880k - 67600k^2$ $= 25$										
Completes a fully correct, logical argument to show that $\text{Var}(5Y - 2) = 25$ AG	AO2.1	R1											

Q2.

Marking Instructions	AO	Marks	Typical Solution
Uses sum of probs = 1	AO1.2	M1	$0.4 + b + c = 1$
Uses formula for $E(R)$	AO1.1a	M1	$b + c = 0.6$
Uses formula for variance $E(X^2) - (E(X))^2$	AO1.1a	M1	$E(R) = 0.2 \Rightarrow$ $(-2 \times 0.3) + (0 \times b) + (a \times c) + (4 \times 0.1) = 0.2$
Obtains a , b and c CAO	AO1.1b	A1	$ac = 0.4$ $E(X^2) - (E(X))^2 = (4 \times 0.3) + (0 \times b) + (a^2 \times c) + (16 \times 0.1) - (0.2)^2 = 3.56$ $a^2c = 0.8$ From (2) and (3) $a = 2$ Hence $c = 0.2$ and $b = 0.4$
Total 4 marks			

Q3.

Marking Instructions	AO	Marks	Typical Solution
Obtains w from $\int_0^{15} w l \, dl = 1$ Allow one error if method correct	AO1.1a	M1	$\int_0^{15} w l \, dl = w \left[\frac{l^2}{2} \right]_0^{15}$ $= \frac{225w}{2} = 1$
Obtains $E(L^2)$ by integrating $\int l^2 \times f(l) \, dl$ FT 'their' value for w	AO1.1a	M1	Hence $w = \frac{2}{225}$ $E(L^2) = \int_0^{15} l^2 \times \frac{2}{225} l \, dl$
Obtains $E(S)$ by evaluating $\frac{1}{15}(0+1+2+3+4+5)$ Allow one error if method correct	AO1.1a	M1	$= \frac{2}{225} \left[\frac{l^4}{4} \right]_0^{15} = \frac{225}{2}$ $E(S) =$
Uses	AO1.1a	M1	

$E(T) = E(I^2) + \frac{1}{2}E(S)$ (PI)			$\frac{1}{15}(0 \times 0 + 1 \times 1 + 2 \times 2 + \dots 5 \times 5)$
$E(T) = 114 \frac{1}{3}$ Shows that Mark awarded if they have a completely correct solution, which is clear, easy to follow and contains no slips AG	AO2.1	R1	$= \frac{55}{15} = \frac{11}{3}$ $E(T) = E(I^2) + \frac{1}{2}E(S)$ $= \frac{225}{2} + \frac{11}{6} = \frac{343}{3} = 114 \frac{1}{3}$ AG
Total 5 marks			

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Identifies that total probability must sum to 1	AO3.1a	M1	$c + 225k - k(15 - 12)^2 = 1$
	Substitutes $t = 12$ to form an equation in c and k .	AO1.1a	M1	$c + 216k = 1$ $1 - c = 216k$
	Rearranges equation with correct rigorous workings to show required result AG	AO2.1	R1	
(ii)	Correct expression for $E(T)$ (PI)	AO1.1a	M1	$E(T) = (-2 \times c) + \int_0^{12} t \times f(t) dt$ $f(t) = \frac{d}{dt}(225k - k(15 - t)^2)$ $= 2k(15 - t)$
	Uses $k = \frac{1}{240}$ and integrates with correct limits	AO1.1a	M1	$E(T) = -0.2 + 2k \int_0^{12} 15t - t^2 dt$ $0.9 = 216k \text{ so } k = \frac{9}{2160} = \frac{1}{240}$ $E(T) = -0.2 + \frac{2}{240} \left[\frac{15}{2}t^2 - \frac{1}{3}t^3 \right]_0^{12}$
	Obtains correct value for $E(T)$	AO1.1b	A1	$= -0.2 + \frac{1}{120} \times 504 = -0.2 + 4.2$ $E(T) = 4$

(b)	Correct expression for $E(\sqrt{ T })$	AO1.1a	M1	$E(\sqrt{ T }) = 0.1 \times \sqrt{2}$ $+ \int_0^{12} 2k \times \sqrt{t} \times (15 - t) dt$ $=$
	Integrates and uses limits	AO1.1a	M1	$\frac{1}{120} \left(10 \times 12^{\frac{3}{2}} - \frac{2}{5} \times 12^{\frac{5}{2}} \right)$ $= \frac{12^{\frac{3}{2}}}{120} \left(10 - \frac{24}{5} \right)$
	Completes clear correct rigorous workings to show required result AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.	AO2.1	R1	$= \frac{12 \times \sqrt{12}}{120} \times \frac{26}{5} = \frac{13\sqrt{12}}{25} = \frac{26\sqrt{3}}{25}$ $E(\sqrt{ T }) = 0.1 \times \sqrt{2} + \frac{26\sqrt{3}}{25}$ $= \frac{\sqrt{2}}{10} + \frac{26\sqrt{3}}{25}$ $= \frac{5\sqrt{2} + 52\sqrt{3}}{50}$
Total 9 marks				

Q5.

(a) (i) $1 - \left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} \right)$
OE

$= \frac{1}{20} = 0.05$
AG

M1

A1

2

(ii) $E(X) =$
 $1 \times \frac{1}{3} + 2 \times \frac{1}{4} + 3 \times \frac{1}{5} + 4 \times \frac{1}{6} + 5 \times \frac{1}{20}$
 At least 2 terms

M1

$= 2.35$
 OE: give B2 for only 2.35 seen

A1

2

- (iii) $E(X^2) =$
 $1 \times \frac{1}{3} + 4 \times \frac{1}{4} + 9 \times \frac{1}{5} + 16 \times \frac{1}{6} + 25 \times \frac{1}{20}$
 $(= 7.05)$
All 5 terms
 $E(X^2) = 7.05$ with no working scores M0
Correct working but labelled Var(X) and then no more done
also scores M0

M1

$\text{Var}(X) = E(X^2) - E(X)^2$
Applied to this problem

m1

$= 1.5275$
 AG

A1

3

- (iv) $1 - (\frac{1}{3} + \frac{1}{4})$ or $(\frac{1}{5} + \frac{1}{6} + \frac{1}{20})$
 $= \frac{5}{12}$ or 0.417
AWRT Accept answer only for B2

M1

A1

2

- (b) $'2.35' \times 100 - 50$
Their value of mean

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$= 185$

FT from (a)(ii)
Give B2 for only 185 seen

A1F

$100^2 \times 1.5275$ or $100 \times \sqrt{1.5275}$

M1

$\text{SD} = \sqrt{15275} = 5\sqrt{611} = 124$
AWFW 123.5 to 124 or $5\sqrt{611}$
Give B2 for only 123.5 to 124 or $5\sqrt{611}$ seen

A1

4

[13]

Q6.

(a)

r	1	2	3	4	5
p	.5	.24	.144	.0864	0.0296

$$0.4 \times 0.6 = 0.24$$

$$0.24 \times 0.6 = 0.144$$

$$0.144 \times 0.6 = 0.0864$$

B1 for any 1 correct (unsimplified) (B1)

B2 all correct and simplified

B2,1

2

(b) P(fewer than 3 bedrooms **are** rented)

$$= P(R = 1, 2) \Rightarrow$$

P(fewer than 3 bedrooms **not** rented)

$$= 1 - P(R = 1, 2)$$

M1

$$= 1 - P(1 \text{ or } 2 \text{ rooms are rented})$$

$$= 1 - (0.5 + \mathbf{0.24})$$

$$[\text{their } 0 < p(2) \leq 0.4704 \text{ value from table used}]$$

$$= 1 - 0.74$$

m1

$$= 0.26$$

A1

Alternative:

P(fewer than 3 not rented)

$$= P(0, 1 \text{ or } 2 \text{ not rented})$$

$$= P(5, 4 \text{ or } 3 \text{ are rented})$$

$$= P(R = 3, 4, 5)$$

M1

$$p = 0.4 \times 0.6^2 + 0.4 \times 0.6^3 + 0.0296$$

$$= \mathbf{0.144} + \mathbf{0.0864} + 0.0296$$

m1

$$[\text{or their } p(3) + p(4) \leq 0.4704 \text{ value from table used}]$$

$$= 0.26 \text{ (cao)}$$

A1

$$[\text{SC } 0.74 \text{ for B1}]$$

3

(c) (i) $E(R) = 0.5 \times 1 + 0.4 \times 0.6 \times 2 + 0.4 \times 0.6^2 \times 3 + 0.4 \times 0.6^3 \times 4 + 0.0296 \times 5$

$$= 0.5 \times 1 + 0.24 \times 2 + 0.144 \times 3 + 0.0864 \times 4 + 0.0296 \times 5$$

$$\sum_{i=1}^5$$

$\times P(R = r_i) \text{ from their table}$

M1

$$= \mathbf{0.5} + \mathbf{0.48} + \mathbf{0.432} + \mathbf{0.3456} + \mathbf{0.148}$$

$$\left[= \frac{1}{2} + \frac{12}{25} + \frac{54}{125} + \frac{216}{625} + \frac{37}{250} \right]$$

$$(0.5 + 1.2576 + 0.148)$$

$$\therefore E(R) = 1.9056$$

$$[awfw 1.9 \text{ to } 1.91] \left[1 \frac{566}{625} \right]$$

A1

2

$$\begin{aligned} \text{(ii)} \quad E(R^2) &= 0.5 \times 1^2 + 0.4 \times 0.6 \times 2^2 \\ &+ 0.4 \times 0.6^2 \times 3^2 + 0.4 \times 0.6^3 \times 4^2 + 0.0296 \times 5^2 \\ &[0.5 + 0.96 + 1.296 + 1.3824 + 0.74] \end{aligned}$$

$$E(R^2) = 4.8784$$

AG

B1

$$\text{Var}(R) = 4.8784 - 1.9056^2 (= 1.24708864)$$

$$4.8784 - \text{their } E^2(R)$$

M1

$$= 1.25(3sf)$$

$$(awfw 1.23 \text{ to } 1.25)$$

A1

3

$$\text{(d)} \quad E(M) = 1250E(R) - 282 = 1250 \times 1.9056 - 282 = 2100$$

cao

B1

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$$\text{Var}(M) = 1250^2 \times [4.8784 - 1.9056^2]$$

$$1250^2 \times \text{their } \text{Var}(R) > 0 \text{ in (c)(ii)}$$

$$(1\,948\,473 \text{ to } 1\,953\,125)$$

$$\text{sd}(M) = 1250 \times \sqrt{1.24708864}$$

$$\text{sd}(M) = \sqrt{1948437} = 1395.9$$

$$(\sqrt{1953125} = 1397.5)$$

$$= 1395.91$$

$$(awfw 1395 \text{ to } 1400)$$

A1

3

[13]

Q7.

(a) (i) $E(X) = \sum_x x \times P(X = x) = 28$

B1

$$E(X^2) = \sum_x x^2 \times P(X = x) = 9$$

$$\text{Var}(X) = 9 - 2.8^2$$

(their $E(X^2)$ – their $E^2(X)$)

M1

$$= 1.16$$

cao

A1

3

(ii) $E(S) = 3 \times E(X) = 8.4$

on their $E(X)$

B1ft

$$\text{Var}(S) = 3 \times \text{Var}(X) = 3.48$$

3 × their $\text{Var}(X)$ from (i) > 0

NB There was a problem with part (a)(ii) which affected the marking of this part.
Please see the Report on the Examination for details.

B1ft

2

(b) $E(Y) = 3.5$

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$$E(Y^2) = 13$$

for $E(Y)$

B1

$$\text{Var}(Y) = 13 - 3.5^2 = 0.75$$

on their $E(Y)$ and $E(Y^2)$

M1

$$\text{Var}(Y) > 0$$

A1ft

$$E(T) = 3 \times E(Y) = 10.5$$

cao

B1

$$\begin{aligned}
 \text{Var}(T) &= 3^2 \times \text{Var}(Y) \\
 &= 9 \times 0.75 \\
 &= 6.75
 \end{aligned}$$

cao

M1

$$= 6.75$$

cao

A1

Alternative:

T	3	6	9	12	
T^2	9	36	81	144	
P	$\frac{1}{20}$	$\frac{2}{20}$	$\frac{3}{20}$	$\frac{4}{20}$	
$E(T)$	$\frac{3}{20}$	$\frac{12}{20}$	$\frac{27}{20}$	$\frac{48}{20}$	10.5
$E(T^2)$	$\frac{9}{20}$	$\frac{72}{20}$	$\frac{243}{20}$	$\frac{576}{20}$	117

(M1A1)

(M1A1)

$$\begin{aligned}
 \text{Var}(T) &= E(T^2) - [E(T)]^2 \\
 &\text{(used)}
 \end{aligned}$$

(M1)

$$\begin{aligned}
 &= 117 - 10.5^2 \\
 &= 6.75
 \end{aligned}$$

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(A1)

6

(c) (i) $P(X > 1) = 0.8$

B1

1

(ii) $P(T = 3) = \frac{1}{20}$ and $P(T = 3 \text{ or } 6) = \frac{3}{20}$

$$P(X + T \leq 9 \text{ and } X > 1)$$

$$= P([2, 3 \text{ or } 6], [3, 3 \text{ or } 6], [4, 3])$$

$$= 0.1 \times \frac{3}{20} + 0.4 \times \frac{3}{20} + 0.3 \times \frac{1}{20}$$

$$= 0.015 + 0.06 + 0.015$$

$$= 0.09$$

Alternative:

$$P(T = 3) = \frac{1}{20} \text{ and } P(T = 6) = \frac{2}{20}$$

$$P(X + T \leq 9 \text{ and } X > 1)$$

$$= P([(2, 3, 4), 3] [(2, 3), 6])$$

$$= 0.8 \times \frac{1}{20} + 0.5 \times \frac{2}{20}$$

$$= 0.04 + 0.05 = 0.09$$

SC

B4

[any 4 correct p's from table B2]

[0.12 **B3**] [0.096 **or** 0.072 **B2**]

$$\begin{cases} P(T = 3) = \frac{1}{20} \text{ and } P(T = 3 \text{ or } 6) = \frac{3}{20} \\ P(T = 3) = \frac{1}{20} \text{ and } P(T = 6) = \frac{2}{20} \end{cases}$$

B1

4

$$(iii) \quad P(X + T \leq 9 \mid X > 1) = \frac{0.09}{0.80}$$

their (c)(ii)

$$\frac{0.09}{0.80} \quad (0 < p < 1)$$

M1

$$= \frac{9}{80} (0.1125)$$

cao

A1

2

[18]

Q8.

$$(a) \quad E(X) = \sum xp$$

$$= \frac{3}{40} + \left(2 \times \frac{6}{40}\right) + \left(3 \times \frac{9}{40}\right) + \left(4 \times \frac{12}{40}\right) + \left(5 \times \frac{5}{20}\right) = 3.5$$

B2,1

2

(b) (i)
$$E\left(\frac{1}{X}\right) = \sum \frac{1}{x} \times p$$

$$= \left(1 \times \frac{3}{40}\right) + \left(\frac{1}{2} \times \frac{6}{40}\right) + \left(\frac{1}{3} \times \frac{9}{40}\right) + \left(\frac{1}{4} \times \frac{12}{40}\right) + \left(\frac{1}{5} \times \frac{5}{20}\right)$$

At least 4 of these terms added (accept decimal equivalents)

M1

$$= \frac{7}{20}$$

AG (allow 0.35 seen)

A1

2

(ii)
$$E\left(\frac{1}{X^2}\right) = \sum \frac{1}{x^2} \times p$$

$$= \left(1 \times \frac{3}{40}\right) + \left(\frac{1}{4} \times \frac{6}{40}\right) + \left(\frac{1}{9} \times \frac{9}{40}\right) + \left(\frac{1}{16} \times \frac{12}{40}\right) + \left(\frac{1}{25} \times \frac{5}{20}\right)$$

At least 4 of these terms added (accept decimal equivalents)

$\frac{133}{800}$
(can be implied by $\frac{133}{800}$ seen with no other working shown)

M1

$$= \frac{133}{800}(0.16625)$$

A1

$$\text{Var}\left(\frac{1}{X}\right) = \frac{133}{800} - \frac{49}{400}$$

$$\left[\text{their } E\left(\frac{1}{X^2}\right) \right] - \left(\frac{7}{20}\right)^2$$

m1

$$= \frac{7}{160}$$

AG (allow 0.04375 seen)

Adep1

4

(c) (i)

x	1	2	3	4	5
y	40	20	13⅓	10	8

p	$\frac{3}{40}$	$\frac{6}{40}$	$\frac{9}{40}$	$\frac{12}{40}$	$\frac{10}{40}$
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Identifying $X = (2), 3, 4, 5$ or

$$Y = (20), 13\frac{1}{3}, 10, 8$$

Alternative:

$$Y < 20 \Rightarrow \frac{40}{X} < 20 \Rightarrow 40 < 20X \Rightarrow X > 2$$

M1 (allow $<$ or $\leq$ **and** $>$ or $\geq$ in above)

M1

$$P(X > 2) = \frac{9}{40} + \frac{12}{40} + \frac{5}{20}$$

$$P(Y < 20) = P(X > 2)$$

A1

$$= P(Y < 20)$$

$$= 1 - \left(\frac{3}{40} + \frac{6}{40} \right) \quad \text{A1}$$

$$= \frac{31}{40} \quad (0.775)$$

$$= \frac{31}{40} \quad (0.775) \quad \text{A1}$$

A1

3

(ii) $\frac{9}{40}$ seen irrespective of labelling

As numerator or final answer (0.225)

B1

$$P(X < 4) \mid Y < 20 = \frac{\frac{9}{40}}{\frac{31}{40}} = \frac{0.225}{0.775}$$

$$= \frac{\frac{9}{40}}{(\text{their}(c)(i))} \quad (\text{or correct use of table})$$

M1

$$= \frac{9}{31} (0.290)$$

AWFW 0.29 to 0.2904

A1

3

[14]

Q9.

(a) (i) $a = \frac{25}{63}$ (OE)
 $\left(\frac{100}{252} \text{ or } \frac{50}{126} \text{ or } 0.397 \right)$

B1

1

(ii) $E(X) = 2.5$ (symmetry)

B1

1

(iii) $E(X^2) = \left(1 \times \frac{25}{252} \right) + \left(4 \times \frac{25}{63} \right) + \left(9 \times \frac{25}{63} \right) + \left(16 \times \frac{25}{252} \right) + \left(25 \times \frac{1}{252} \right)$

$\sum x^2 \times p$ attempted

M1

$E(X^2) = \frac{125}{18}$
 $\left(6 \frac{17}{18} \text{ or } 6.94 \right)$

A1

$\text{Var}(X) = \frac{125}{18} - \frac{25}{4}$
 $[(\text{their } E(X^2)) - (\text{their } E(X))^2]$

$\text{dep } \sum x^2 \times p$ used

m1

$= \frac{25}{36}$
 $0.694 \quad [\text{Var} > 0]$

A1

$\text{sd}(X) = \frac{5}{6}$
 $0.833 \left(\sqrt{(\text{their } \text{Var}(X))} \right) (\text{dep m1})$

A1ft

5

(b) (i) $E(\text{Pay}) = \frac{4}{9} \times 90$ pence
= 40 pence

M1

$\Rightarrow$ Joanne expected to make a loss
(loss of 10p per game)

A1

Alternative:

$$\frac{5}{9} > \frac{2}{9} + \frac{2}{9} \Rightarrow \text{loss (for B1) then M1A1}$$

(ii) $E(\text{Loss}) = 100 \times 10$ pence
= £10
 $100 \times (\text{their loss/game})$

B1ft

3

[10]

Q10.

(a) (i) $P(R \geq 5) = 0.3 + 0.25 + 0.1 + 0.05$
= 0.70

CAO

B1

1

(ii) $E(R) = \sum rp$
= $3 \times 0.1 + 4 \times 0.2 + 5 \times 0.3 + 6 \times 0.25 + 7 \times 0.1 + 8 \times 0.05$

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= 5.2

A1

2

(iii) $E(R^2) = 9 \times 0.1 + 16 \times 0.2 + 25 \times 0.3 + 36 \times 0.25$
+ $49 \times 0.1 + 64 \times 0.05$
(Correct expression **or** 28.7)

M1A1

(= 28.7)

$\text{Var}(R) = 28.7 - 5.2^2$

M1

= 1.66

AG

A1

4

(b) (i) $P(R + S = 6) = 0.1 \times 0.15 = 0.015$
 $P(R = 3 \text{ and } S \leq 5) = 0.1 \times 0.85 = 0.085$

B1

$$\begin{aligned} P(R + S = 7) &= 0.1 \times 0.4 + 0.2 \times 0.15 \\ &= 0.04 + 0.03 \\ &= 0.07 \\ P(R = 4 \text{ and } S \leq 4) &= 0.2 \times 0.55 = 0.110 \end{aligned}$$

B1

$$\begin{aligned} P(R + S = 8) &= 0.2 \times 0.4 + 0.1 \times 0.3 + 0.3 \times 0.15 \\ &= 0.08 + 0.03 + 0.045 \\ &= 0.155 \end{aligned}$$

$$P(R = 5 \text{ and } S \leq 3) = 0.3 \times 0.15 = 0.045$$

B1

$$\begin{aligned} P(R + S \leq 8) &= 0.015 + 0.07 + 0.155 \\ P(R + S \leq 8) &= 0.085 + 0.110 + 0.045 \end{aligned}$$

M1

$$\begin{aligned} &= 0.24 \\ &= 0.24 \text{ (AG)} \end{aligned}$$

A1

5

(ii) $p = {}^5C_4(0.24)^4(0.76) + (0.24)^5$
 $= 0.0126 + 0.000796$
First term correct
+ correct second term
or correct numerical values
(must use $p = 0.24$)

M1M1

$$\begin{aligned} &= 0.0134 \\ &\quad (0.013 \text{ to } 0.0135) \end{aligned}$$

A1

3

(iii) $P(R = 4 \mid R + S \leq 8)$
 $= \frac{P(R = 4 \text{ and } R + S \leq 8)}{P(R + S \leq 8)}$

Alternative: (using (b)(i))

$$= \frac{P(R=4 \text{ and } S \leq 4)}{P(R+S \leq 8)}$$

$$= \frac{0.11}{0.24} = \frac{11}{24}$$

$$= \frac{0.03 + 0.08}{0.24}$$

(numerator) or 0.11 seen

B1

(÷ 24 iff numerator < 0.24)

M1

$$= \frac{11}{24} (0.458)$$

CAO

A1

3

[18]

Q11.

(a)

r	1	2	3	4
$P(R=r)$	$\frac{2}{3}$	$\frac{2}{9}$	$\frac{2}{27}$	k

$$k + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} = 1 \Rightarrow k = \frac{1}{27}$$

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M1

AG

A1

2

(b) $P(R \geq 3) = \frac{2}{27} + \frac{1}{27} = \frac{1}{9}$

Allow $\frac{3}{27}$ or 0.111

B1

1

(c) (i) $C = 27R + 5$

$$E(R) = \left(1 \times \frac{2}{3}\right) + \left(2 \times \frac{2}{9}\right) + \left(3 \times \frac{2}{27}\right) + \left(4 \times \frac{1}{27}\right)$$

$$= 1\frac{13}{27}$$

$$(1.48) \text{ or } \frac{40}{27}$$

B1

$$\therefore E(C) = 27 \times 1\frac{13}{27} + 5$$

M1

$$= 45$$

A1F

3

$$(ii) \quad E(R^2) = \left(1 \times \frac{2}{3}\right) + \left(4 \times \frac{2}{9}\right) + \left(9 \times \frac{2}{27}\right) + \left(16 \times \frac{1}{27}\right)$$

$$= 2\frac{22}{27} \text{ or } \frac{76}{27}$$

$$(2.81)$$

B1

$$\text{Var}(R) = 2\frac{22}{27} - \left(1\frac{13}{27}\right)^2$$

M1

$$= \frac{452}{729}$$

$$(0.62)$$

$$\therefore \text{St. dev}^n(C) = 27 \times \sqrt{\frac{452}{729}}$$

$$27 \times \sqrt{\text{Var}(R)} [\text{Var}(R) > 0]$$

M1

$$= 21.3$$

$$\text{CAO (21.26)}$$

A1

4

$$\text{SC: Var}(C) = 452 \text{ (CAO)}$$

$$(B1M1B1A0)$$

Alternative (c)

C	32	59	86	113
P	$\frac{2}{3}$	$\frac{2}{9}$	$\frac{2}{27}$	$\frac{1}{27}$

C	32	59	86	113
n	18	6	2	1

$$\bar{x} = 45 \text{ and } \sigma = 21.260$$

from calculator

$$\left(\bar{x} = \frac{\sum Cn}{27} \right)$$

[10]

Q12.

(a) (i) $P(\text{GG or YY or RR})$

M1

$$= \frac{2}{10} \times \frac{1}{9} + \frac{3}{10} \times \frac{2}{9} + \frac{4}{10} \times \frac{3}{9}$$

$$= \frac{2}{9}$$

(AG)

A1

(ii) $P(\text{BB or BB}) = \frac{1}{10} \times \frac{9}{9} + \frac{9}{10} \times \frac{1}{9}$

$$= \frac{1}{10} + \frac{9}{10} \times \frac{1}{9}$$

M1

$$= \frac{1}{5}$$

(AG)

A1

2

(b) (i)

	Same	1 Blue	Neither
x	135	145	-45

$P(X = x)$	$\frac{2}{9}$	$\frac{1}{5}$	$\frac{26}{45}$
------------	---------------	---------------	-----------------

B1B1

2

(ii) $E(X) = 135 \times \frac{2}{9} + 145 \times \frac{1}{5} + (-45) \times \frac{26}{45}$

Multiply two rows of their table from (b)(i)

M1

$$= 29 + 30 - 26 = 33 \text{ pence}$$

AG

A1

2

(c) (i) $E(Y) = 104 - 3E(X)$

$$= 104 - 3 \times 33 = 5 \text{ pence}$$

$\therefore$ Joanne would expect to win £5

OE (eg 500p)

M1

A1

A1

3

(ii) $E(X^2) = 9425$
 $(4205 + 4050 + 1170)$

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B1

$$\text{Var}(X) = 9425 - 33^2 = \mathbf{8336}$$

$$sd(X) = 91.30$$

$$\text{Var}(Y) = 9 \times \text{Var}(X)$$

B1

$$= 9 \times 8336$$

$$9 \times (\text{their } \text{Var}(X) > 0)$$

$$= 75024$$

$$\text{or } 3 \times (\text{their } sd(X))$$

M1

$\Rightarrow$ standard deviation (Y) = 274 pence

273.9p or £2.74

A1

4

[15]

Q13.

(a) $P(X \geq 5) = P(X = 5) + P(X = 6)$
 $= \frac{5}{20} + \frac{6}{24}$

M1

$$= \frac{1}{2}$$

A1

2

(b) (i) $E\left(\frac{1}{X}\right) = \sum \frac{1}{x} \times P(X = x) =$
 $\left(1 \times \frac{1}{20}\right) + \left(\frac{1}{2} \times \frac{2}{20}\right) + \left(\frac{1}{3} \times \frac{3}{20}\right) + \left(\frac{1}{4} \times \frac{4}{20}\right) + \left(\frac{1}{5} \times \frac{5}{20}\right) + \left(\frac{1}{6} \times \frac{6}{24}\right)$
Use of $\sum \frac{1}{x} \times p$

M1

$$= \frac{1}{4} + \frac{1}{24}$$

$$= \frac{7}{24}$$

AG

A1

2

(ii) $E\left(\frac{1}{X^2}\right) = \frac{109}{900}$
Use of $\sum \frac{1}{x^2} \times p$

M1

or 0.21

A1

$$\text{Var}\left(\frac{1}{X}\right) = \frac{109}{900} - \left(\frac{7}{24}\right)^2$$

AG

A1

$$= 0.036 \text{ (3dp)}$$

3

(c) $A = \frac{1}{X}(X + 3)$

$$A = 1 + \frac{3}{X}$$

(either)

B1

$$E(A) = 1 + 3E(X^{-1})$$

$$E(A) = 1 + 3 \times \frac{7}{24}$$

(either)

M1

$$= 1 \frac{7}{8}$$

(1.875)

A1

$$\text{Var}(A) = \text{Var}\left(1 + \frac{3}{X}\right) = 9\text{Var}(X^{-1})$$

M1

$$= 9 \times \frac{173}{4800}$$

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allow 9×0.036

$$= 0.324 \text{ or } \frac{519}{1600}$$

0.324375

A1

5

[12]

Q14.

(a) (i) $E(Y) = \sum y P(Y = y)$
 $= 5 \times 0.1 + 15 \times 0.2 + 25 \times 0.3 + 35 \times 0.4$
 $= 25$

B1

$$\begin{aligned}\text{Var}(Y) &= E(Y^2) - [E(Y)]^2 \\ &= 725 - 25^2\end{aligned}$$

M1

$$\begin{aligned}&= 100 \\ &\text{CAO}\end{aligned}$$

A1

$$\begin{aligned}\text{Standard deviation} &= 10 \\ &\text{ft on } \text{Var}(Y) > 0\end{aligned}$$

A1ft

$$\begin{aligned}\text{(ii)} \quad C &= 10Y + 5 \\ E(C) &= 10E(Y) + 5 \\ &= 10 \times 25 + 5 \\ &= 255 \text{ pence} \\ &\text{OE}\end{aligned}$$

B1

1

$$\begin{aligned}\text{(b)} \quad \text{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= 75.25 - 8.35^2\end{aligned}$$

M1

$$\begin{aligned}&= 75.25 - 69.7225 \\ &= 5.5275\end{aligned}$$

AWFW 5.52 to 5.53

A1

$$\begin{aligned}T &= 0.4X + 250 \\ \text{Var}(T) &= \text{Var}(0.4X + 250) \\ &= 0.4^2 \times \text{Var}(X) \\ &\text{Var}(X) > 0\end{aligned}$$

M1

$$\begin{aligned}&= 0.16 \times 5.5275 \\ &= 0.8844\end{aligned}$$

AWFW 0.884 to 0.885

A1

4

[9]

Q15.

$$\text{(a)} \quad \Sigma p = 1$$

$$k = 1 - (0.40 + 0.25 + 0.18 + 0.12)$$

$$k = 0.05$$

B1

1

(b) (i) $E(X) = \sum_{\text{all } x} x P(X = x) = 3.17$

B1

1

(ii)
$$\text{Var}(X) = \sum_{\text{all } x} x^2 P(X = x) - \mu^2$$

$$\sum x^2 P(X = x) \text{ attempted}$$

M1

$$= 11.53 - 10.0489$$

$$\sum x^2 P(X = x) - [E(X)]^2$$

M1

$$= 1.4811$$

$$= 1.48$$

$$\text{AWFW } 1.48 - 1.49$$

A1

3

(c) (i) $E(Y) = 2E(X) - 3$

$$= 2 \times 3.17 - 3$$

$$= 3.34$$

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M1

1

(ii) $\text{Var}(Y) = 4 \text{Var}(X)$

their $\text{Var}(X) \times 4$

M1

$$= 4 \times 1.48$$

$$= 5.92$$

$$\text{Standard deviation} = 2.43$$

$$\sqrt{\quad}$$

Note: if separate table used for Y:

$$E(Y^2) = 17.08$$

$$[E(Y)]^2 = 11.16$$

m1A1

3

[9]

Q16.

- (a) $\Sigma p = 1$
 $k = 1 - (0.01 + 0.05 + 0.14 + 0.30 + 0.12)$
 $k = 0.38$

B1

1

- (b) (i) $E(X) = \sum_{\text{all } x} x P(X = x) = 4.35$
 $\frac{87}{20}$

B1

1

- (ii) $\text{Var}(X) = \sum_{\text{all } x} x^2 P(X = x) - \mu^2$

E(X²) attempted

M1

$$= 20.09 - 18.9225$$

$$\sum x^2 P(X = x) - \mu^2$$

M1

$$= 1.1675$$

$$\frac{467}{400} \text{ (AWFW 1.16 - 1.17)}$$

A1

3

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- (c) (i) $E(Y) = 5E(X) + 2$
 $= 5 \times 4.35 + 2$
 $= 23.75$
Their (b)(i) $\times 5 + 2$

M1

1

- (ii) $\text{Var}(Y) = 25\text{Var}(X)$
 $= 29.1875$
Their (b)(ii) $\times 25$

M1

$$\text{Standard deviation} = 5.40$$

$$\sqrt{\quad}$$

m1

$$(5.40 - 5.41)$$

A1

3

[9]

Q17.

(a) $1 - 3p \geq 0$ and $p \geq 0$

M1

$$3p \leq 1$$

A1

$$0 \leq p \leq \frac{1}{3}$$

A1

3

(b) (i) $E(X) = (1 \times p) + (2 \times p) + (3 \times p) + 4(1 - 3p)$
 $= 4 - 6p$

B1

1

(ii) $E(X^2) = (1 \times p) + (4 \times p) + (9 \times p) + 16(1 - 3p)$
 $= 16 - 34p$

Attempting $E(X^2)$

M1

$$\text{Var}(X) = (16 - 34p) - (4 - 6p)^2$$

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M1

$$= 16 - 34p - 16 + 48p - 36p^2$$

$$= 14p - 36p^2$$

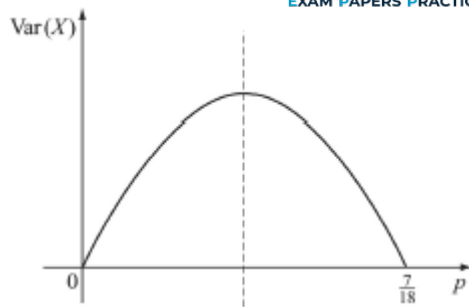
$$= 2p(7 - 18p)$$

AG

A1

3

(c) (i) Max value of quadratic graph occurs at:



Or

$$\frac{dV}{dp} = 14 - 17p \quad \text{and} \quad \frac{d^2V}{dp^2} = -72$$

$$\text{For max, } V, \frac{dV}{dp} = 0 \quad \text{and} \quad \frac{d^2V}{dp^2} < 0$$

M1

$$p = \frac{1}{2} \left(0 + \frac{7}{18} \right) = \frac{7}{36}$$

$$\Rightarrow p = \frac{7}{36}$$

A1

2

(ii) Max value of $\text{Var}(X)$

$$= 2 \times \frac{7}{36} \times \left(7 - 18 \times \frac{7}{36} \right)$$

Attempted

M1

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$$= 2 \times \frac{7}{36} \times \frac{7}{2}$$

$$= \frac{49}{36}$$

$$\text{sd}_{\max}(X) = \frac{7}{6} (1.17)$$

(square root attempted)

M1

CAO

A1

3

[12]

Q18.

(a) (i)

x	4	-1
$P(X = x)$	$\frac{1}{5}$	$\frac{4}{5}$

B1

1

(ii) $E(X) = \left(4 \times \frac{1}{5}\right) + \left(-1 \times \frac{4}{5}\right) = 0$
 $(p > 0, \sum p = 1)$

M1A1

2

(b)

x	4	-1
$P(X = x)$	$\frac{1}{3}$	$\frac{2}{3}$

B1

$$E(X) = \left(4 \times \frac{1}{3}\right) + \left(-1 \times \frac{2}{3}\right) = \frac{2}{3}$$

$$(p > 0, \sum p = 1)$$

B1

$$E(24X) = 24 \times E(X)$$

$$= 24 \times \frac{2}{3}$$

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$$= 16$$

A1

4

[7]

Q19.

(a) $E(X) = \sum_{\text{all } x} x^2 P(X = x)$

$$= 50$$

CAO

B1

$$E(X^2) = \sum_{\text{all } x} x^2 P(X = x)$$

$$= 2602.6(0)$$

M1

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= 2602.6 - 50^2$$

$$= 102.6(0)$$

M1

$\Rightarrow$ standard deviation $(X) = 10.13$
(to nearest 1p)

A1

4

(b) $E(Y) = \mu = E(10X + 250)$

$$= 10 \times E(X) + 250$$

$$= 750$$

On their $E(X)$

B1ft

$$\text{s.d.}(Y) = 10 \times 10.1$$

$$= 101$$

On their $\text{sd}(X)$

B1ft

EXAM PAPERS PRACTICE² [6]

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Q20.

(a) $\mu = E(X) = \sum_{all} x P(X = x)$

$$\begin{aligned} \mu &= (1 \times 0.09) + (2 \times 0.13) + (3 \times 0.50) \\ &\quad + (4 \times 0.15) + (5 \times 0.12) \end{aligned}$$

M1

$$\mu = 3.05$$

A1

$$\begin{aligned} \sigma^2 &= \text{Var}(X) = E(X^2) - \mu^2 \\ &= 10.51 - 3.05^2 \end{aligned}$$

M1

$$= 1.2075$$

$$\therefore \sigma = \sqrt{1.2075} = 1.10$$

$$(1.099)$$

A1

4

(b) $P(\mu - \sigma < X < \mu + \sigma)$

$$= P(1.95 < X < 4.15)$$

M1

$$= P(2 \leq X \leq 4)$$

M1

$$= 0.78$$

A1

3

[7]

Q21.

$$E(R) = \sum_{\text{all } r} r P(R=r)$$

(a) $= \left(1 \times \frac{7}{16}\right) + \left(2 \times \frac{5}{16}\right) + \left(3 \times \frac{3}{16}\right) + \left(4 \times \frac{1}{16}\right)$

$$= \frac{30}{16}$$

$$= 1\frac{7}{8}$$

$$(1.875)$$

B1

$$E(R^2) = \sum_{\text{all } r} r^2 P(R=r)$$

$$= \frac{70}{16} \text{ or } 4\frac{3}{8}$$

$$(4.375)$$

B1

$$\begin{aligned}\text{Var}(R) &= 4 \frac{3}{8} - \left(1 \frac{7}{8}\right)^2 \\ &= \frac{220}{256} \text{ or } \frac{55}{64}\end{aligned}$$

(0.859375)

A1

4

(b) (i) $32 \times \frac{1}{4} = 8$

1

(ii)
$$\begin{aligned}&= \left(32 \times \frac{7}{16} \times \frac{1}{5}\right) + \left(32 \times \frac{5}{16} \times \frac{1}{2}\right) + 8 \times \frac{9}{10} \\ &= 2.8 + 5 + 7.2 \\ &= 15\end{aligned}$$

M1

A0 if these numbers rounded before adding

A1

2

[7]

Q22.

(a)
$$E(R) = \left(1 \times \frac{1}{4}\right) + \left(2 \times \frac{1}{2}\right) + \left(4 \times \frac{1}{4}\right)$$

M1A1

$$= 2.25$$

$$E(R^2) = \left(1 \times \frac{1}{4}\right) + \left(4 \times \frac{1}{2}\right) + \left(16 \times \frac{1}{4}\right)$$

$$= 6.25$$

$$\therefore \text{Var}(R) = 6.25 - (2.25)^2$$

M1

$$= 1.1875$$

$$1\overline{16}^3 \text{ on their } E(R)$$

A1ft

4

(b) (i)

x	1	$\frac{1}{4}$	$\frac{1}{16}$
$P(X = x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

B1

$$E(X) = (1 \times \frac{1}{4}) + (\frac{1}{4} \times \frac{1}{2}) + (\frac{1}{16} \times \frac{1}{4})$$

M1

$$= \frac{1}{4} + \frac{1}{8} + \frac{1}{64}$$

$$= \frac{16 + 8 + 1}{64}$$

$$= \frac{25}{64}$$

AG

EXAM PAPERS PRACTICE

A1

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3

$$(ii) \quad A = \left(R + \frac{8}{R}\right) \times \frac{8}{R} = 8 + \frac{64}{R^2}$$

Attempt at area = $l \times b$

M1

$$E(A) = 8 + E\left(\frac{64}{R^2}\right) = 8 + 64 \times E\left(\frac{1}{R^2}\right)$$

M1

$$= 8 + 64 \times E(X) = 8 + 64 \times \frac{25}{64}$$

$= 33$
CAO

A1
3
[10]

Q23.

(a) $1(.1) + 2(.4) + 3(.4) + 4(.1) = 2.5$
OE; AG

E1
1

(b) $E(X^2) = 6.9$
PI by full correct calculation

$V(X) = 6.9 - 2.5^2 = 0.65$
NMS 3/3

B1
M1A1
3
[4]

Q24.

(a) $k = \frac{1}{8}$

B1
1

(b) (i) Method for mean
Award even if candidate then divides, eg by 5

M1

Mean = 3.5
ft wrong value for k (<1)

A1F
2

(ii) $E(X^2) = 16.25$
PI by full correct calculation; ft as above

B1F

Var (X) = $16.25 - 3.5^2 = 4$
ft wrong values

M1A1F

Q25.

- (a) Method for mean

Allow even if c then divides, eg by 5

M1

Mean = 0.6

NMS 2/2

A1

2

- (b) $E(X^2) = 2$

PI; award even if this is c's variance

B1

Variance = $2 - 0.6^2 = 1.64$

ft one wrong value; NMS 3/3

M1A1F

3

[5]

Q26.

- (a) $r:$ 0 1 2 3 4

$P(R = r)$ 0.1 0.2 0.4 0.2 0.1

- (i) $E(R) = 0 + 0.2 + 0.8 + 0.6 + 0.4 = 2$

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AG; use of $\sum r \times p_r$ or symmetrical argument

M1

1

- (ii) $E(R^2) = 0 + 0.2 + 1.6 + 1.8 + 1.6 = 5.2$

CAO; must be some evidence of use of $\sum r^2 \times p_r$

B1

$\text{Var}(R) = E(R^2) - (E(R))^2 = 1.2$

AG; use of a formula for $\text{Var}(R)$

M1

or

$= 0.4 + 0.2 + 0 + 0.2 + 0.4 = 1.2$

CAO; ≥ 4 terms correct

(B1)

2

(b) $E(P) = 3 \times 2 + 4 = 10$
CAO

B1

$\text{Var}(P) = 3^2 \times \text{Var}(R)$
Use of $\text{Var}(aX + b) = a^2\text{Var}(X)$
with $a > 1$ and $b \geq 0$

M1

$= 10.8$

CAO

A1

3

(c) (i) $C = 200 - R - P$
Use of; may be implied

M1

$= 200 - R - (3R + 4)$
Hence $C = 196 - 4R$

CAO

A1

2

(ii) $E(C) = 196 - 4 \times 2 = 188$
CAO

B1

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$\text{Var}(C) = 4^2 \times \text{Var}(R) = 19.2$
CAO; dependent on A1 in (c)(i)

B1dep

2

[10]

Q27.

Mean = 3

Allow NMS

B1

$E(X^2) = \frac{31}{3}$

PI; allow AWRT 10.3 even if this is c's variance

B1

$$\text{Variance} = \frac{31}{3} - 3^2 = \frac{4}{3}$$

*Allow AWRT 1.33; NMS 3/3;
f.t. wrong value for mean or $E(X^2)$*

M1A1F

4

[4]

Q28.

- (a) Area of rectangle is given by:

$$A = S \times \left(2 + \frac{40}{S}\right) = 2S + 40$$

cao; may be implied by $E(A) = 50$

B1

$$\text{Mean: } E(A) = 2 \times 5 + 40 = 50$$

cao

B1

$$\text{Variance: } \text{Var}(A) = 2^2 \times \text{Var}(S)$$

*use of $\text{Var}(aX + b) = a^2 \text{Var}(X)$
with $a > 1$ and $b \geq 0$*

M1

$$= 4 \times 33 = 132$$

cao

A1

4

- (b) (i) $\begin{matrix} S: & 1 & 5 & 10 & 20 \\ t = 40/S: & 40 & 8 & 4 & 2 \\ p: & 0.5 & 0.3 & 0.1 & 0.1 \end{matrix}$

$$\begin{matrix} t = 40/S: & 40 & 8 & 4 & 2 \\ p: & 0.5 & 0.3 & 0.1 & 0.1 \end{matrix}$$

cao or equivalent

B1

$$E(T) = \sum t \times P(S = s) = \sum t \times p$$

use of $\sum x \times P(X = x)$

M1

$$\begin{aligned} &= 40 \times 0.5 + 8 \times 0.3 + 4 \times 0.1 + 2 \times 0.1 \\ &= 20 + 2.4 + 0.4 + 0.2 \\ &= 23 \end{aligned}$$

cao or equivalent

AG

A1

- (ii) Perimeter of rectangle is given by:
 $P = 2 \times (S + (2 + T)) = 2S + 2T + 4$
 Mean: $E(P) = 2 \times 5 + 2 \times 23 + 4 = 60$
 cao

B2

2

[9]

Q29.

Mean is 4.6

Allow NMS

B1

$$E(X^2) = 25.2$$

PI

B1

$$\text{Variance} = 25.2 - 4.6^2$$

Allow M1 even if c then takes the square root

M1

$$= 4.04$$

NMS 3/3; ft wrong value for mean or $E(X^2)$

A1F

4

EXAM PAPERS PRACTICE [4]

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Q30.

- (a) $\frac{1}{4}, \frac{1}{4}$ inserted in table

Accept statement, or use of correct probs in (b)

B1

1

- (b) Method for mean

Award M1 A0 if c then divides,
 e.g. by 6

M1

$$\text{Mean} = 4$$

Allow NMS

A1

$$E(X^2) = 19$$

PI; allow even if c gives this as variance.

B1

$$\text{Variance} = 19 - 4^2 = 3$$

*Alternative: $\sum p(x - 4)^2 = 3$ M1 A1
(-1 EE); Allow NMS*

M1 A1

[6]

Q31.

(a)

$v:$	1	2	3	4	5
$p:$	0.15	0.45	0.20	0.15	0.05

$$E(V) = \sum vp = 2.5$$

Evidence of $\sum vp$ used; AG

M1

$$E(V^2) = \sum v^2 p = 7.4$$

CAO or implied correctly

B1

$$\therefore \text{Var}(V) = E(V^2) - (E(V))^2$$

Evidence of $E(V^2) - (E(V))^2$ used

M1

$$= 7.4 - 2.5^2 = 1.15$$

AG

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(b) $E(C) = (30 \times 2.5) + 25 = 100$

CAO

B1

$$\text{Var}(C) = 30^2 \text{Var}(V)$$

$$= \text{Var}(aX + b) = a^2 \text{Var}(X)$$

M1

$$= 1035$$

CAO

A1

3

(c) $E(R) = 40E(V^2) + 15E(V) + 50$

use of, OE

M1

$$= (40 \times 7.4) + (15 \times 2.5) + 50$$

f.t. on $E(V^2)$, or B1 for 105, 240, 455, 750, 1125

A1f.t.

$$= 383.5$$

CAO
NB ignore units in (b) & (c)

A1

3

[9]

Q32.

(a) (i) $E(M_1 + M_2 + M_3) = 3 \times 1.7 = 5.1$

B1

1

(ii) $\text{Var}(M_1 + M_2 + M_3) = 3 \times 0.81 = 2.43$

B1

1

(b) (i) $E(R) = \sum p \times R$
 $= 2 \times \frac{1}{8} + 4 \times \frac{9}{16} + 6 \times \frac{5}{16}$

M1

$$= 4\frac{3}{8}$$

4.375 (AWRT 4.38)

A1

$$\text{Var}(R) = \sum p \times R^2 - E^2(R)$$

$$= 4 \times \frac{1}{8} + 16 \times \frac{9}{16} + 36 \times \frac{5}{16} - \left(4\frac{3}{8}\right)^2$$

M1

$$= 20\frac{3}{4} - 19\frac{9}{64}$$

$$= 1\frac{39}{64}$$

1.609375 (AWRT 1.61)

A1

4

$$(ii) \quad E\left(\frac{1}{2}R\right) = \frac{1}{2} E(R) = \frac{1}{2} \times 4\frac{3}{8} = 2\frac{3}{16}$$

B1f.t.

$$\text{Var}\left(\frac{1}{2}R\right) = \frac{1}{4} \times \text{Var}(R)$$

$$= \frac{1}{4} \times 1\frac{39}{64}$$

M1

$$= \frac{103}{256}$$

0.40234 (AWRT 0.402)
(on their (b)(i) Var(R) value)

A1f.t.

3

$$(c) \quad (i) \quad T = \frac{1}{2}R + M_1 + M_2 + M_3$$

$$E(T) = E\left(\frac{1}{2}R\right) + E(M_1 + M_2 + M_3)$$

$$= 2\frac{3}{16} + 5.1$$

$$= 7.29$$

(AWRT 7.29)

7.2875 (a)(i) + (b)(ii)

$$(if (b)(ii) \quad E\left(\frac{1}{2}R\right) = \frac{1}{2} E(R) \text{ used.})$$

B1f.t.

1

$$(ii) \quad \text{Var}(T) = \text{Var}\left(\frac{1}{2}R\right) + \text{Var}(M_1 + M_2 + M_3)$$

$$= \frac{103}{256} + 2.43$$

(their a(ii) + b(ii))

M1

$$= 2.832$$

(AWRT 2.83)

A1f.t.

2

[12]

Q33.

- (a) Method for mean

Award M1 even if c then divides, eg by 4

M1

$$\text{Mean} = 3.4$$

Allow NMS 2/2

A1

2

- (b) $E(X^2) = 12.4$

seen or implied

B1

Method for variance of RV

M1

$$\text{Variance} = 12.4 - 3.4^2 = 0.84$$

Allow NMS 3/3; ft one wrong

A1F

3

[5]

Q34.

- (a) $E(R) = 3$ $\text{Var}(R) = 1.2$
 $E(5(2R - 1)) = E(10R - 5) = 25$
 CAO

B1

$$\text{Var}(10R - 5) = 10^2 \text{Var}(R)$$

use of Var (aX ± b) = a²Var(X) with a > 1 and b ≥ 0

M1

$$= 120$$

CAO

A1

3

- (b) (i) $\frac{60}{r} : 60 \quad 30 \quad 20 \quad 15 \quad 2$

B1



$p:$ 0.1 0.2 0.4 0.2 0.1

CAO; OE; must be stated*

CAO; must be stated; dep on previous B1*

(*)not from (ii) unless tables

B1

2

$$(ii) \quad E\left(\frac{60}{R}\right) = \sum \frac{60}{r} \times p$$

use of values of p must be correct

M1

1

$$= 24.2$$

AG

$$(iii) \quad \text{Var}\left(\frac{60}{R}\right) = E\left(\frac{3600}{R^2}\right) - \left(\frac{60}{R}\right)^2$$

$$= 759.4 - 24.2^2$$

use of

M1

$$= 173.7 \text{ to } 173.8$$

AWFW

exact answer exact answer is 173.76

do not accept 173 or 174

A1

2

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[8]

Q35.

(a) All digits are equally likely

So

by symmetry

$$P(D = d) = \frac{1}{9}$$

B1

So

$$E(D) = \frac{45}{9} = 5$$

B1

2

(b) $\text{Var}(D) = E(D^2) - (E(D))^2$

Use of

M1

$$= \frac{95}{3} - 5^2 = \frac{20}{3} \text{ or } 6\frac{2}{3}$$

CAO

Decimal must clearly indicate $\frac{2}{3}$

A1

2

(c) $N = 3(D + 5)$
 $E(N) = 3 \times 5 + 15 = 30$
 CAO

B1

$$\text{Var}(N) = 3^2 \times \text{Var}(D)$$

Use of $\text{Var}(aX + b) = a^2\text{Var}(X)$
 with $a > 1$ and $b \geq 0$

$$= 9 \times \frac{20}{3} = 60$$

CAO

M1

A1

3

[7]

Q36.

(a) $\frac{1}{6}$ ©2025 Exam Papers Practice. All rights reserved.
 allow 0.17

B1

1

(b) $1 \times \frac{1}{2} + 2 \times \frac{1}{3} + 3 \times \frac{1}{6} = \frac{5}{3}$ A.G

B1

4

$$E(X^2) = \frac{1}{2} + \frac{4}{3} + \frac{9}{6} \left(= \frac{10}{3} \right)$$

f.t. on part(a) seen

M1A1f.t.

$$\text{Variance} = \frac{5}{9} \text{ or } 0.56$$

<i>no working</i>	0.55	SR1
0.56	SR2	
$\frac{5}{9}$	3	

A1

[5]

Q37.

(a) $C = 2\pi R$

accept 2π or πd OE

B1

$$\therefore E(C) = 2\pi \times E(R) = 4\pi$$

CAO

B1

$$\therefore \text{Var}(C) = (2\pi)^2 \times \text{Var}(R)$$

Use of $V(aX) = a^2V(X)$ with $a > 1$

M1

$$= 4\pi^2$$

CAO or AWRT 39.5

A1

4

(b) $A = \pi R^2$

CAO; award only if 1st B1 in (a) not scored

(B1)

$$\therefore E(A) = \pi \times E(R^2)$$

Use of $E(aX^2) = aE(X^2)$ with $a > 1$

M1

$$= 5\pi$$

CAO or AWRT 15.7

A1

2

[6]

Q38.

Method for mean

Allow even if candidate then divides inappropriately

M1

Mean is 1.7

NMS $\frac{1}{2}$

A1

Formula for variance of RV

Allow even if formula not used

M1

$E(N^2) = 4.1$

OE

A1

Variance = $4.1 - 1.7^2 = 1.21$

NMS 1/3; ft wrong value for $E(N^2)$ or mean

A1F

5

[5]

Q39.

Mean = $2\frac{1}{2}$

Allow NMS

B1

Either

$E(X^2) = 7$

PI; Allow even if c gives this as variance

B1

Variance = $7 - \left(\frac{5}{2}\right)^2$

Allow if either or both divided by 4

M1

$\dots = \frac{3}{4}$

ft one wrong value

A1F

OR

$$\text{Variance} = \sum p(x - \bar{x})^2$$

$$= \frac{3}{4}$$

-1 EE

(A2,1)

4

[4]

Q40.

(a) (i) $E(X) = 1.00 \times 0.35 + 1.50 \times 0.21 + 2.00 \times 0.24 + 3.00 \times 0.20 = £1.745$

M1

(ii) $E(X^2) = 1.00^2 \times 0.35 + 1.50^2 \times 0.21 + 2.00^2 \times 0.24 + 3.00^2 \times 0.20 =$

$$3.5825$$

$$3.58 \sim 3.59$$

M1

A1

(iii) standard deviation = $\sqrt{3.5825 - 1.745^2} =$

*for variance (but not if called standard deviation)
for standard deviation*

M1

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$$£0.733$$

$$0.73 \sim 0.734$$

allow 0.537 (0.537 ~ 0.538) if called variance

M1 A1

6

(b) (i) $0.24 + 0.20 = 0.44$
cao

M1 A1

(ii) mode = 1

B1

Probability $\geq$ mode = 1
cao

B1

4

(c) (i) decrease –

B1

children under 5 are charged the least (zero).

B1

(ii) increase –

B1

spread of distribution increased.

B1

4

[14]

Q41.

(a) (i) $E(X) = 0 \times 0.22 + 1 \times 0.25 + 2 \times 0.2 + 3 \times 0.14 + 4 \times 0.19$
 $= 1.83$

method

1.83 cao

M1

A1

(ii) $E(X^2) = 0 \times 0.22 + 1 \times 0.25 + 4 \times 0.2 + 9 \times 0.14 + 16 \times 0.19$

method

$= 5.35$

M1

(iii) $V(X) = 5.35 - 1.83^2 = 2.001$

method for variance – may be calculated as $E[(X - E(X))^2]$

disallow if called standard deviation

M1

s.d. = $\sqrt{2.0011}$
 $\sqrt{\text{variance}}$

m1

$= 1.41$

1.41 - 1.42

allow for variance = 2.00 (2.00 - 2.01)

A1

(b) (i) $P(X < 3) = 0.22 + 0.25 + 0.2 = 0.67$
0.67 cao

B1

(ii) $P(X > 1.83) = 0.2 + 0.14 + 0.19$
'their' mean
requires correct method for mean

m1

$= 0.53$
0.53 cao

A1

(iii) median = 2
2 cao

$P(X > 2) = 0.14 + 0.19 = 0.33$
0.33 cao

B1

B1

5

[11]

Examiner reports

Q1.

The question was attempted well by most students with many scoring full marks. Some students lost the final mark by writing the correct calculations but not making clear what the calculations were finding. Other students lost the final mark by writing X when referring to the random variable Y . A minority of students mistook the random variable as continuous when calculating expectation and variance whilst treating it as discrete when finding k . Some students simply added the probabilities when attempting to find the expectation and found the expectation when attempting to find the variance. Very few students attempted the question by finding the values of $5Y - 2$ first and most of these were unsuccessful.

Q5.

The majority of students scored very well on this question, although a few treated X as a continuous variable and performed various integrations and so scored few, if any, marks. Students should have appreciated that, in part (a)(iii), where they were required to 'Show that $\text{Var}(X) = 1.5275$ ', simply quoting values such as $E(X^2) = 7.05$ without any working, was not acceptable. This 'general principal' can be expected to apply on future papers. Part (b) was well done, although a small proportion of students, having found the variance, did not proceed to the standard deviation.

Q6.

This was the best answered question on the paper with many fully correct solutions seen. Almost all students managed to complete the table in part (a) correctly. Weaker students found part (b) very difficult to fathom. However, there were very many fully correct solutions with most making good use of the probabilities found in part (a). Part (c) was especially well done with many correct solutions to all parts. However, some students lost credit for using premature approximation or for failing to write down the value of the standard deviation having correctly found the value for the variance.

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Q7.

In part (a)(i), candidates usually managed to find the correct values for $E(X)$ and $\text{Var}(X)$. Part (a)(ii) was considered to be invalid, as described at the beginning of this report. In part (b), there were many excellent solutions seen. Unfortunately, as is usual when the probability distribution is defined using formulae, several candidates thought the distribution to be continuous and consequently used integration methods in their attempts at $E(T)$ and $\text{Var}(T)$ for no reward.

The majority of candidates wrote down the correct value of 0.8 as their answer to part (c)(i). Part (c)(ii) was only answered well by the most able of candidates, and so the correct answer of 0.09 was very rarely seen. Many candidates attempted to find the correct probabilities with some success. However, they then multiplied their answer by 0.8 (or the value they had obtained in part (c)(i)), which rather negated the good work that they had done up to that point. Probably the most efficient ways of doing this part of the question were either to evaluate

$P(X = 2, 3 \text{ or } 4) \times P(T = 3 \text{ or } 6) - P(X = 4 \text{ and } T = 6)$ as $0.8 \times 0.15 - 0.01 \times 0.3 = 0.09$ or to complete a table of the required probabilities as shown below.

	$X = 2$	$X = 3$	$X = 4$
$T = 3$ (0.05)	$X + T = 5$ 0.05×0.1 $= 0.005$	$X + T = 6$ 0.05×0.4 $= 0.02$	$X + T = 7$ 0.05×0.3 $= 0.015$
$T = 6$ (0.1)	$X + T = 8$ 0.1×0.1 $= 0.01$	$X + T = 9$ 0.1×0.4 $= 0.04$	$X + T = 10$ 0.1×0.3 $= 0.03$

Some candidates included $X = 1$ and consequently gained an answer of 0.12, which some then again multiplied by 0.8 to obtain the very popular incorrect answer of 0.096. Only the most able candidates were able to answer part (c)(iii) correctly.

Q8.

It was very disappointing to see some candidates still using integration methods in this question involving discrete random variables. There were several attempts at

$$\int xP(X = x)dx \quad \text{seen in part}$$

(a) and $\int \frac{1}{x} P(X = x) dx$ in part (b)(i). However, part (a) was very well done by the majority of candidates. In parts (b)(i) and (ii), where the answers were given, many candidates failed to show sufficient method to warrant full marks.

Candidates should have indicated in some way that they had used $\sum \frac{1}{x} \times p$ and $\sum \frac{1}{x^2} \times p$ somewhere in their calculations.

This was not often the case although the majority of candidates ended up with the correct given answers, often following dubious or incorrect methods.

The responses to part (c)(i) were very disappointing. Only the best candidates seemed capable of identifying the values of $Y < 20$ and then adding their corresponding probabilities.

Part(c)(ii) was found to be the hardest part on the paper.

Very few candidates realised that $P(X < 4 \mid Y < 20)$ equated to $P(X = 3) = \frac{9}{40}$.

Consequently, although some candidates realised that they had to divide 'something' by their answer to part (c)(i), they gained no marks.

Q9.

The great majority of candidates gave the correct answer to part (a)(i), though a small minority stated that $a = 100$. In part (a)(ii), although most candidates gave the correct answer, not many were able to *write down* the value of $E(X)$ by looking at the symmetry of

the distribution; the majority of candidates used $E(X) = \sum_{\text{all } x} x \times P(X = x)$ to find the correct answer. Part (a)(iii) was done well with very many correct answers seen.

However, some candidates, having found that $\text{Var}(X) = \frac{25}{36}$, failed to go on to calculate the standard deviation. Some of those candidates, who used decimals rather than

fractions, occasionally ended up with $\text{Var}(X) = 0.69$ giving $\text{sd}(X) = 0.831$ and as a consequence, lost an accuracy mark. It is emphasised that, as final answers are normally required to be given correct to 3 significant figures unless otherwise stated, working throughout should be to at least 4 significant figures.

Part (b) of this question was not well done by the majority of candidates. The most popular misconception, which gained no marks, was that Joanne made a loss because $P(\text{Loss}) = \frac{7}{9} > P(\text{Win}) = \frac{2}{9}$. In fact $P(\text{Loss}) = \frac{5}{9} > P(\text{Win}) = \frac{4}{9}$ which lead to the conclusion that Joanne could expect to make a loss.

When the correct reason for a loss was given, candidates were often then unable to

calculate the magnitude of the loss with $\frac{5}{9} \times 100 \times 10 - \frac{4}{9} \times 100 \times 90 = -1222 \frac{2}{9}$ often seen.

Candidates should have calculated $\frac{5}{9} \times 100 \times 50 - \frac{4}{9} \times 100 \times 40 = 1000\text{p}$ (£10) or

$50 \times 100 - \frac{4}{9} \times 100 \times 90 = 1000\text{p}$ (£10). A considerable number of candidates' answers

were penalised by 1 mark for rounding $\frac{5}{9} \times 100$ to 56 and $\frac{4}{9} \times 100$ to 44. This led to $56 \times 50 - 44 \times 40 = 1040\text{p}$ (£10.40) as the amount lost by Joanne.

Q10.

Part (a) was usually answered well. However, in part (a)(iii), too many candidates did not show a sufficiently complete and accurate method to gain full marks.

There were many correct solutions seen to part (b). In part (b)(i), those candidates who seemed to have very little or no idea simply attempted to multiply the corresponding values in the two tables together and then wrote the given answer of 0.24; this gained no credit.

In part (b)(ii), most candidates realised that binomial probabilities were required, but unfortunately candidates often only considered one term. Even when two terms were attempted, the incorrect expression $0.24^4 \times 0.76 + 0.24^5$ was often seen.

Part (b)(iii) was only answered correctly by the most able candidates. Some found $P(R = 4 \text{ and } S \leq 4)$ and thought that this was the required answer, whilst others realised that they had to divide $P(R = 4 \text{ and } R + S \leq 8)$ by 0.24 but were unable to find the correct value for the numerator.

Q11.

Far too many candidates treated the given distribution as *continuous* rather than *discrete* and consequently attempted to apply integration methods throughout. This obviously met with no success but also wasted a lot of time. Part (a) was usually done quite well. However, it must be pointed out that when an answer is given in the question, then an appropriate method must be shown by candidates in order to gain full credit. Simply evaluating $P(R = r)$ for $r = 1, 2, 3$ and then writing down the given answer was not

sufficient; some mention of $\sum_{\text{all } r} P(R = r) = 1$ had to be seen. $P(R = 4) =$

$\frac{2}{3} \left(\frac{1}{3} \right)^{4-1} = \frac{1}{27} = k$ was also seen but again gained no marks.

In part (b), some candidates incorrectly interpreted 'at least 3' as either meaning 'greater than 3' or 'less than or equal to 3' or even 'equal to 3', with $\frac{1}{27}$, $\frac{1}{26}$ and $\frac{2}{27}$ as common wrong answers.

There were very many correct solutions seen to parts (c)(i) and (c)(ii). The most common cause for loss of marks was in part (c)(ii), where a number of candidates correctly found $\text{Var}(C) = 452$ but then did not go on to attempt to find the standard deviation of C as requested. Several candidates confused the use of R and C .

Q12.

This was often the worst answered question on the paper. Many candidates were unable to show enough convincing working in part (a) to gain full credit. However, part (b) was usually done quite well, except by those candidates who simply wrote down a 2×2 table or stated +45 rather than -45. Most realised that the probabilities had to add up to one, so

correctly obtaining $\frac{26}{45}$.

The values in the table were then usually used correctly to obtain the given value of $E(X)$. Part (c)(i) was usually done quite well with many finding $E(Y) = 5$. Unfortunately some candidates failed to multiply this answer by 100 to give the final answer of 500 pence or £5. There were some excellent solutions, by various methods, to the final part of this question, but these were mainly given by the more able candidates.

Q13.

Although it was stated clearly in the question that X was a discrete random variable, many candidates employed integration methods throughout this question, all to no avail. Those who did treat the variable as discrete usually did very well, with many gaining full marks. Although there were many fully correct answers of 0.5 seen to part (a), there were several candidates who, having lost both marks here, went on to gain all other available marks for the question.

In part (b), where the answers were given, it was very pleasing to see candidates usually supplying the required valid working to enable them to gain full credit.

Unfortunately, there were still a few candidates who calculated $E(X) = \frac{17}{4}$ and then

wrongly assumed that $E\left(\frac{1}{X}\right) = \frac{1}{E(X)}$. A similar incorrect argument followed when

attempting to find the value of $\text{Var}\left(\frac{1}{X}\right)$.

Part (c) was very well done, even by some of those candidates who failed to gain any credit on the previous parts of this question.

Q14.

This question proved to be a good source of marks for most candidates. In part (a)(i), the great majority gained full marks. However, there was a minority who, having found correctly $\text{Var}(Y) = 100$, failed to identify the standard deviation. Part (a)(ii) was done well by most candidates with $E(C) = 255$ often seen. In part (b), the majority of candidates found the value for $\text{Var}(X)$ and then went on to use $\text{Var}(T) = 0.4^2 \times \text{Var}(X)$ to gain the correct answer of 0.884. Those candidates who attempted to use $\text{Var}(T) = E(T^2) - [E(T)]^2$, a valid if not the most efficient method, correctly found $E(T) = 253.34$ but were then unable to show that $E(T^2) = E(0.16X^2 + 200X + 62500) = 64182.04$. As a result they often ended up with a negative answer for the variance.

Q15.

Candidates found this question very much to their liking, with the vast majority gaining full marks. Most candidates easily obtained 0.05 for the value of k in part (a), before using this value to help to obtain the correct answer of 3.17 for $E(X)$ in part (b)(i). Almost all

candidates managed to calculate $E(X^2) = \sum x^2 \times P(X = x)$ in part (b)(ii), with very few wrongly thinking that this was the same as the variance. In part (c), although the majority of candidates used the most efficient ways of finding $E(Y)$ and $\text{SD}(Y)$, by evaluating $E(Y) = 2 \times E(X) - 3$ and $\text{SD}(Y) = \sqrt{4 \times \text{Var}(X)} = 2\sqrt{\text{Var}(X)}$, there were frequent attempts made to tabulate the probability distribution for Y followed by the use of methods already employed in part (b). Although this process was often successful, it used up valuable time which could have been allocated to other more demanding parts of the paper.

Q16.

This proved to be a very popular question, with the overwhelming majority of candidates gaining full marks. Parts (a) and (b)(i) were done well, with very few numerical errors

seen. Almost all candidates managed to calculate $E(X^2) = \sum x^2 \times P(X = x)$ in part (b)(ii), with very few candidates wrongly thinking that this was the same as the variance.

In part (c), although the majority of candidates used the most efficient ways of finding $E(Y)$ and $\text{SD}(Y)$ by evaluating $E(Y) = 5 \times E(X) + 2$ and $\text{SD}(Y) = \sqrt{25 \times \text{Var}(X)} = 5 \times \sqrt{\text{Var}(X)}$, there were frequent attempts made to tabulate the probability distribution for Y , followed by the use of methods already employed in part (b). Although this process was often successful, it used up valuable time which could have been allocated to other more demanding parts of the paper. A few confused candidates thought that $\text{Var}(Y) = 25 \times E(X)$, whilst others simply found the variance, not realising that the stand requested.

Q17.

This was the worst-answered question on the paper with the vast majority of candidates gaining fewer than half marks. The fact that probability values must fall within the range $0 \leq p \leq 1$ implied that $0 \leq p < 1$ by inspection of the first three values in the table. Inspection

of the fourth value gave $1 - 3p \geq 0$, implying that $p \leq \frac{1}{3}$. This was not usually explained well by the weaker candidates. In part (b), $E(X)$ and $E(X)^2$ were usually attempted correctly with many fully correct expressions in terms of p seen. The majority of candidates then attempted to use the correct method by stating and then using $\text{Var}(X) = E(X^2) - (E(X))^2$. Unfortunately, a lack of good algebraic skills let some candidates down. They were either unable to cope with $(4 - 6p)^2$, where $16 - 36p^2$ was a common wrong expression, or they could not cope with the negative signs when attempting subtractions.

Thus, it was hardly surprising that the correct answer was usually seen even though the working did not really lead to the correct expression given on the question paper for $\text{Var}(X)$.

Part (c)(i) asked for the value of p which gave a maximum value of $\text{Var}(X)$. Differentiation methods were employed by candidates when tackling this part. In such cases it should be

$$\frac{dV}{dp} = 0 \text{ and } \frac{d^2V}{dp^2} < 0$$

remembered that it is then expected that both should be considered. This was not always seen. No candidate considered properties of a quadratic in order to

gain the correct answer of $p = \frac{7}{36}$. In part (c)(ii), in most cases where candidates managed to find a value for the variance, they rightly went on to attempt to deduce a value for the standard deviation.

Q18.

The majority of candidates found this question to be quite straightforward with many scoring full marks in part (a). A few mistakes were seen in part (b), usually from those candidates who confused the 'total mark' with the 'total number of questions' to then quote

an incorrect answer of $\frac{16}{24}$.

Q19.

This was the best answered question on the paper. In part (a), candidates usually found $E(X) = 50$, with most going on to show the correct value of the standard deviation to be 10.13. In part (b), many correct answers were seen, with most candidates using their knowledge of the mean and variance of $(aX + b)$ to show that $E(Y) = 750$ and that the required value of the standard deviation was 101.3. A minority of candidates constructed a new table for the distribution of Y , and then repeated methods used in part (a), often with considerable or complete success.

Q20.

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Most candidates made a very good attempt at part (a), especially in finding the mean of the given distribution. The previously common errors of leaving the answer as σ^2 when σ is

asked for, or using $\text{Var}(X) = \sum x^2 p$ were not usually seen. In part (b), the majority of candidates could arrive at $P(1.95 < X < 4.15)$ but some were then unsure as to how to continue, not realising that this was equivalent to finding $P(2 < X < 4) = 0.78$.

Q21.

In part (a), although the vast majority of candidates were able to calculate the mean and variance of R , many failed to comply with the instruction 'calculate exact values', often writing their answers as rounded decimals. The correct answer of 8 was often seen in part (b)(i) but far too many candidates either misinterpreted 'at least 3' and thus calculated

$\frac{3}{16} \times 32 = 6$, or in some cases $\frac{15}{16} \times 32 = 30$. A minority of candidates simply wrote $\frac{1}{4}$ as

the answer instead of evaluating $\frac{4}{16} \times 32$. Weaker candidates usually just calculated 90% of 32 giving them an answer of 28.8. There were many good attempts at part (b)(ii)

with the correct answer of 15 often seen.

Q22.

Statistics 2A

This question also proved to be a valuable source of marks for many candidates. The vast majority of them performed well on part (a) with only a few losing credit for not giving an **exact** answer as asked for in the question. In part (b)(i), the table of the probability distribution of X , although asked for in the question, was conspicuous by its absence. Candidates should realise that failure to answer the question as set will inevitably result in a loss of marks. A good number made excellent attempts at part (b)(ii) with the majority able to find an expression for the area followed by the correct answer of 33 for $E(A)$.

Statistics 2B

This question again proved to be a valuable source of marks for many candidates. The vast majority of them performed well on part (a) with only a few losing credit for not giving an exact answer as asked for in the question. In part (b)(i), the table of the probability distribution of X , although asked for in the question, was conspicuous by its absence. Candidates should realise that failure to answer the question as set will inevitably result in a loss of marks. A good number of candidates made excellent attempts at part (b)(ii) with the majority able to find an expression for the area followed by the correct answer of 33 for $E(A)$.

Q23.

This question was well answered by the great majority of candidates. In part (a) the given answer deterred most (but not all) of the weaker candidates from carrying out an unwanted division by 4, but there was a significant minority who ignored the probabilities

$$\frac{1+2+3+4}{4}$$

altogether and obtained the printed answer from the calculation $\frac{1+2+3+4}{4}$. In part (b) the usual fault occurred of squaring the probabilities instead of (or as well as) the values of X .

Q24.

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The usual errors occurred in this question. Weak candidates sometimes failed to find the correct value for the missing probability, while others ignored the probabilities altogether when attempting to find the mean and variance. Some gave the variance as 16.25, while others subtracted the mean value of X rather than the square of this value.

Q25.

Most candidates were completely successful in this standard question on random variables. There were candidates who went through the process of finding the mean but then divided by 5. Similarly there were some who found $E(X^2)$ but then presented this as the variance. A further error arose from the fact that two of the values of x were negative: many candidates obtained negative values for x^2 .

Q26.

Most candidates scored the first six marks for correct answers to parts (a) and (b). However, some incorrect methods occurred in part (a) with the calculation of $\text{Var}(R)$ as

$\sum (r \times P(R=r))^2$ which, perchance, equalled 1.2. Correct attempts at part (c) were much rarer and generally only the more able candidates achieved significant success. In part (i), simple algebraic manipulation proved to be a major stumbling block. In part (ii), some candidates obtained 188 for $E(C)$ directly from parts (a)(i) and (b). However, the mark for $\text{Var}(C)$ was only available to the small proportion of the candidates who scored two marks in part (i). Some of these candidates then failed to obtain 19.2, apparently thrown by $(-4R)$.

Q27.

This very straightforward test of the mean and variance of a random variable was very well answered by the majority of the candidates. The usual errors occurred: working out the mean and then finding it necessary to divide the result by 4; thinking that the expected value of X^2 was the variance; or alternatively subtracting the mean, rather than the square of the mean, to find the variance.

Q28.

Many candidates obtained $E(A) = 50$ but frequently by somewhat dubious methods. Such candidates rarely presented a correct method for $\text{Var}(A)$; this was left to those candidates who stated initially that $A = 2S + 40$.

Answers to part (b)(i) showed a great improvement on those to similar questions on previous papers with full marks the most common outcome.

In general it was left to the stronger candidates to score the two marks in part (b)(ii) as others either made no attempt or got into somewhat lengthy incorrect calculations.

Q29.

This question was very well answered by almost all the candidates. Only rarely did a candidate divide by 4 in finding the mean, or fail to square the mean as required in the calculation of the variance.

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Q30.

This question was very well answered, with many candidates gaining full marks.

A common error in finding the mean was to divide the correct answer by 6. In calculating the variance, a substantial minority of candidates squared the probabilities instead of, or as well as, the values of the random variable. Another common mistake was to subtract the mean, rather than the square of the mean.

Q31.

Most candidates scored the 3 marks in part (a) without resorting to fiddling answers, but a small minority expressed $E(V^2)$ as $\sum v \times (P(V=v))^2$.

In part (b), whilst most candidates found $E(C)$ correctly, many candidates lost the 2 marks for $\text{Var}(C)$ by either using 30 rather than 30^2 or by adding 25.

Part (c) proved much more difficult, with approximately half the candidates scoring no marks. By far the most common error was to substitute 2.5^2 for $E(V^2)$ rather than 7.4 as

found in part (a). A small number of candidates chose to set up the probability distribution for R and then find $E(R)$ from first principles. Few candidates had complete success using this valid alternative approach.

Q32.

The answers seen to this question were much improved on those seen to similar questions on previous papers.

In part (a)(i) most candidates gained the correct answer of 5.1. Although there were many correct answers of 2.43 seen in answers to part (a)(ii), there was a considerable number of candidates who thought that the required variance could be found by using $\text{Var}(M_1 + M_2 + M_3) = 3^2 \times \text{Var}(M)$.

Most candidates calculated $E(R)$ and $\text{Var}(R)$ correctly with only a few thinking that

$\text{Var}(R) = E(R^2)$. There were many correct solutions to part (b)(ii), with $E\left(\frac{1}{2}R\right) = \frac{1}{2}E(R)$

and $\text{Var}\left(\frac{1}{2}R\right) = \frac{1}{4}\text{Var}(R)$ often seen. However, some candidates ignored the 'hence' and

used the distribution for $\frac{1}{2}R$ together with methods already employed in part (b)(i) in order to find the answers.

The vast majority of candidates knew what was expected of them in part (c) and many fully correct answers were seen.

Q33.

This question was generally well answered. In part (a), a common error was to perform the correct calculation but then to divide this by 4.

In part (b), some candidates squared the probabilities instead of, or as well as, the values of the random variable. A more common mistake was to subtract the mean, rather than the square of the mean.

Q34.

Whilst full marks for answers to this type of question were more common than on previous papers, the majority of candidates scored less than half marks with the weaker candidates often scoring at most one mark. In part (a), most candidates found the correct value for the mean though, sadly, $5(2R - 1) = 10R - 1$ was not uncommon. In attempting to find the variance, candidates used $5(2\text{Var}(R) - 1)$ or multipliers for $\text{Var}(R)$ of $(5^2 \times 2)$ or (5×2^2) rather than 10^2 . A minority of candidates attempted to work from first principles but rarely with much success. Many candidates simply omitted part (b)(i) whilst others wrote down

values of $\frac{60}{P(R=r)}$. Those candidates who wrote down the correct probability distribution, then scored the mark in part (b)(ii). Surprisingly, a small number of candidates recovered the work in part (b)(i) in answering part (b)(ii). Earlier problems within the question rarely deterred candidates from attempting, often successfully, the final part, though not squaring 24.2 before subtraction from 759.4 was not unusual.

Q35.

In answering part (a), most candidates read 'explain' as 'show' and calculated $\frac{\sum d_i}{9}$ or $\sum d_i p_i$ with all $p_i = \frac{1}{9}$ due to random selection. Such answers usually scored the two marks available. Answers to part (b) were generally correct with most candidates offering the answer $\frac{20}{3}$ or $6\frac{2}{3}$ though a minority of candidates subtracted 5 rather than 5^2 . Answers to part (c) often failed to score marks through incorrect algebra, for example $3(E(D) + 5) = 3E(D) + 5$; or through incorrect application of the Var operator, for example $\text{Var}(N) = 3\text{Var}(D)$ or $3\text{Var}(D) + 5$ or even $3E(D) + 5$.

Q36.

Most candidates showed a good grasp of the mean but variance was less well understood.

A common error in part (b) was to use $\sum xp(x)^2$, or $\sum (xpx)^2$, instead of $\sum x^2p(x)$.

Q37.

A large number of candidates appeared unaware of or had forgotten the formulae for the circumference (C) and area (A) of a circle, or interchanged them. Of those who did know, most were able to find the value for $E(C)$ but many then failed to square both 2 and π when finding $\text{Var}(C)$. In part (b), where attempted, the common error was to state that $E(A) = E(\pi R^2) = \pi(E(R)^2) = 4\pi$. It was apparent that this topic in the specification was poorly understood by many candidates.

Q38.

Full marks were often earned in this question. A common mistake made by candidates was to divide by 5, or by some other number, in the calculation of the mean and to make a similar mistake in the evaluation of the variance.

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Q39.

The majority of candidates answered this question correctly and efficiently. No candidate used the symmetry of the distribution to find the mean. A substantial minority calculated the value of $E(X^2)$ but gave this answer as the variance.

Q40.

There was a good spread of marks on this question. Apart from candidates who confused variance with standard deviation or with $E(X)^2$, most were successful with part (a). In part (b) a few incorrectly identified the mode as 0.35. The most common error in part (c) was to say that since there was no charge for under 5s, the mean and standard deviation would be unchanged.

Q41.

This question was well answered. A few candidates thought that $E(X^2)$ and the variance of X were identical. Others confused variance and standard deviation. A number correctly identified and evaluated the variance but did not complete the calculation of the standard

deviation. Some candidates were unable to identify the median and so unable to make progress in part (b)(iii). A few candidates used a normal distribution in part (b).



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